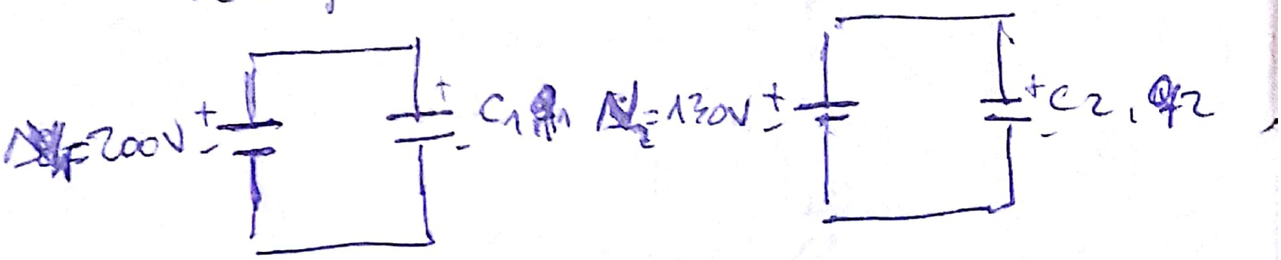
T.P. 5

P.7

Se cargan ambos capacitores



$C_1 = 5\,\mu F$

$\Delta V_1 = 200V$

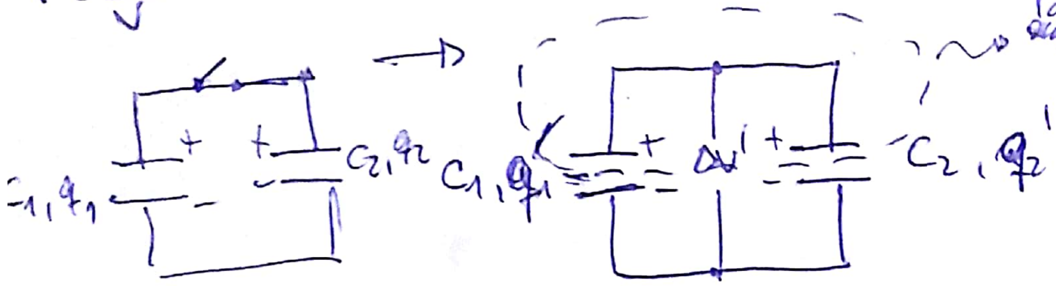
$C_2 = 2\,\mu F$

$\Delta V_2 = 130V$

$F = \frac{C}{V}$

$q_1 = C_1 \Delta V_1 = 5 \cdot 10^{-6} F \cdot 200V = 1\,mC$

$q_2 = C_2 \Delta V_2 = 2 \cdot 10^{-6} F \cdot 130V = 0{,}26\,mC$



$q_{total} = q_1 + q_2 = q_1' + q_2' = \underline{1{,}26\,mC}$

$q_{total} = q_1' + q_2' = C_1 \Delta V' + C_2 \Delta V' = (C_1 + C_2)\Delta V'$

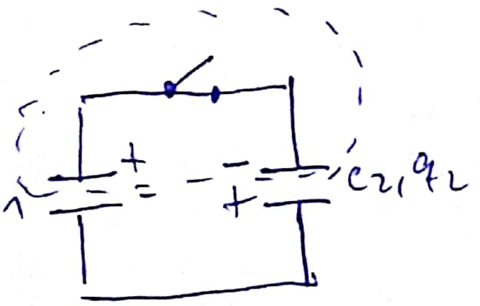
$q_{total} = (C_1 + C_2)\Delta V' = C_{eq} \cdot \Delta V'$

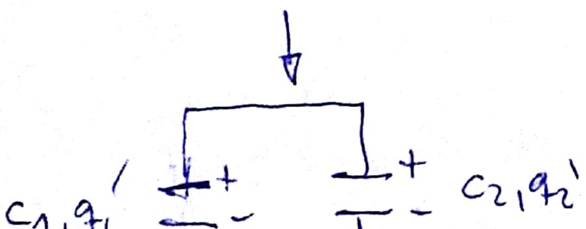
$\Delta V' = \frac{q_{total}}{C_1 + C_2} = \frac{1{,}26 \cdot 10^{-3} C}{5 \cdot 10^{-6} F + 2 \cdot 10^{-6} F} = \underline{180\,V}$

$q_1' = C_1 \Delta V' = 5 \cdot 10^{-6} F \cdot 180V = 0{,}9\,mC$

$q_2' = C_2 \Delta V' = 2 \cdot 10^{-6} F \cdot 180V = 0{,}36\,mC$

b)





$q_{total} = q_1 - q_2 = q_1' + q_2' = 0{,}74\,mC$

$q_{total} = q_1' + q_2' = C_1 \Delta V' + C_2 \Delta V'$

$q_{total} = (C_1 + C_2)\Delta V'$

$\Delta V' = \frac{q_{total}}{C_1 + C_2} = \frac{0{,}74 \cdot 10^{-3} C}{7 \cdot 10^{-6} F} = \underline{105{,}7\,V}$

$q_1' = C_1 \Delta V' = 5 \cdot 10^{-6} F \cdot 105{,}7V = \underline{0{,}528\,mC}$

$q_2' = C_2 \Delta V' = 2 \cdot 10^{-6} F \cdot 105{,}7V = \underline{0{,}211\,mC}$

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